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Vellore Institute of Technology
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School of Computer Science Engineering and Information Systems

Department of Computer Applications

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PAMCA698J – Dissertation -1

0th Review

Register Number	25MCA0261
Student Name	GOKUL S
Dissertation Domain	Cyber Security with Blockchain and Artificial Intelligence for Secure Healthcare Systems
Dissertation Title	AI-Driven Blockchain Framework for Secure Medical Image Sharing with Adaptive Chaos Encryption, Smart Access Control, and AI-Based Tamper Localization
Abstract	The rapid adoption of digital healthcare systems has significantly improved the storage and sharing of medical images such as MRI, CT scan, X-ray, and ultrasound images. However, conventional cloud-based medical image sharing systems face critical challenges, including unauthorized access, data tampering, privacy breaches, and lack of reliable integrity verification. To address these issues, this research proposes an AI-Driven Blockchain Framework for Secure Medical Image Sharing with Adaptive Chaos Encryption, Smart Access Control, and AI-Based Tamper Localization. In the proposed framework, medical images are encrypted using adaptive chaos-based encryption before being stored in the cloud, while cryptographic hashes are securely recorded on a blockchain to ensure data integrity and immutability. Smart contracts are employed to provide secure, role-based access control for authorized healthcare professionals. Additionally, an artificial intelligence module analyzes retrieved images to detect and localize any tampered regions, enhancing trust and reliability in medical image sharing. The framework also maintains an immutable audit trail of all transactions, enabling transparency and forensic investigation when required. The proposed system integrates blockchain, artificial intelligence, cryptography, and cloud computing into a unified healthcare security framework. It is

	expected to improve confidentiality, integrity, authentication, and secure collaboration among hospitals while reducing the risks associated with cyberattacks and unauthorized image manipulation. The proposed solution is suitable for real-world healthcare environments and provides a strong foundation for future research in secure medical image management.
Keywords	Blockchain, Cyber Security, Medical Image Sharing, Artificial Intelligence, Adaptive Chaos Encryption, Smart Contracts, Cloud Computing, Tamper Detection, Data Integrity, Healthcare Informatics.
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